lab13.py - C:/Users/Deekshith/AppData/Local/Programs/Python/Python314/lab13.py (3.14.2)

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```python
import pandas as pd

# DataFrames Creation


emp = pd.DataFrame({
    'ID': [101, 102, 103, 104, 105],
    'Name': ['A', 'B', 'C', 'D', 'E'],
    'Dept': ['HR', 'Eng', 'Eng', 'HR', 'Mkt']
})

sal = pd.DataFrame({
    'ID': [101, 102, 103, 104, 105],
    'Sal': [50000, 70000, 80000, 55000, 60000]
})

sales1 = pd.DataFrame({
    'Date': pd.date_range(start='2026-01-01', periods=5),  # this fun, genetrate the date
    'Reg': ['N']*5,  # same as ['N','N','N','N','N']
    'Sale': [250, 300, 200, 400, 350]
})

sales2 = pd.DataFrame({
    'Date': pd.date_range(start='2026-01-01', periods=5),
    'Reg': ['S']*5,
    'Sale': [300, 320, 280, 360, 310]
})

print("Employee Data:")
print(emp)

print("\nSalary Data:")
print(sal)
```

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```
print(sal)


# Merging
merge_data = pd.merge(emp, sal, on='ID')

print("\nMerged Data:")
print(merge_data)

# GroupBy
avg = emp.merge(sal, on='ID').groupby('Dept')['Sal'].mean() .reset_index()   # as_index=False
avg.columns=['dept','avgsal']

print("\nAverage Salary by Dept:")
print(avg)


# Joining

                    #emp1 = emp[['ID', 'Name', 'Dept']].set_index('ID')
                    #sal1 = sal[['ID', 'Sal']].set_index('ID')
emp1 = emp.set_index('ID')
sal1 = sal.set_index('ID')
join_data = emp1.join(sal1,how='inner')  # default left join

print("\nJoined Data:")
print(join_data)


# Concatenation
final = pd.concat([sales1, sales2], axis=1) # axis=0 vertical(row) , axis=1 horizontal(col)

print("\nConcatenated Sales Data:")
print(final)
```

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Deekshith/AppData/Local/Programs/Python/Python314/lab13.py =
Employee Data:
   ID Name Dept
0  101    A   HR
1  102    B  Eng
2  103    C  Eng
3  104    D   HR
4  105    E  Mkt

Salary Data:
   ID    Sal
0  101  50000
1  102  70000
2  103  80000
3  104  55000
4  105  60000

Merged Data:
   ID Name Dept    Sal
0  101    A   HR  50000
1  102    B  Eng  70000
2  103    C  Eng  80000
3  104    D   HR  55000
4  105    E  Mkt  60000

Average Salary by Dept:
  dept   avgsal
0  Eng  75000.0
1   HR  52500.0
2  Mkt  60000.0
```

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```
2  103  60000
3  104  55000
4  105  60000

Merged Data:
   ID Name Dept   Sal
0  101    A   HR  50000
1  102    B  Eng  70000
2  103    C  Eng  80000
3  104    D   HR  55000
4  105    E  Mkt  60000

Average Salary by Dept:
  dept   avgsal
0  Eng  75000.0
1   HR  52500.0
2  Mkt  60000.0

Joined Data:
   Name Dept   Sal
ID
101    A   HR  50000
102    B  Eng  70000
103    C  Eng  80000
104    D   HR  55000
105    E  Mkt  60000

Concatenated Sales Data:
      Date Reg  Sale      Date Reg  Sale
0 2026-01-01   N   250 2026-01-01   S   300
1 2026-01-02   N   300 2026-01-02   S   320
2 2026-01-03   N   200 2026-01-03   S   280
3 2026-01-04   N   400 2026-01-04   S   360
4 2026-01-05   N   350 2026-01-05   S   310
>>>
```